

Slate Bandits

- At each $t \in [T]$, the learner is presented with N item-sets $\{\mathcal{X}_t^1, \dots, \mathcal{X}_t^N\} \subset \mathbb{R}^d$.
- The learner chooses items $\mathbf{x}_t^i \in \mathcal{X}_t^i$ and plays the N -tuple $\mathbf{x}_t = (\mathbf{x}_t^1, \dots, \mathbf{x}_t^N)$.
- She receives reward $r_t(\mathbf{x}_t)$ sampled from a GLM, i.e, $r_t(\mathbf{x}_t) \mid \mathbf{x}_t$ is distributed as $\mathbb{P}_{\boldsymbol{\theta}}(r_t(\mathbf{x}_t) \mid \mathbf{x}_t) = \exp(r_t(\mathbf{x}_t) \cdot (\mathbf{x}_t^\top \boldsymbol{\theta}^*) - b(\mathbf{x}_t^\top \boldsymbol{\theta}^*) + c(r_t(\mathbf{x}_t)))$.
The link function μ is defined as $\mu(\mathbf{x}_t^\top \boldsymbol{\theta}^*) := \mathbb{E}[r_t(\mathbf{x}_t) \mid \mathbf{x}_t] = b'(\mathbf{x}_t^\top \boldsymbol{\theta}^*)$.
- Here, $\mathbf{x}_t$ is called a *slate* and the item $\mathbf{x}_t^i$ is said to be placed in *slot* i .
- The learner aims to choose *slates* in a way to minimize her (expected) regret,

$$R(T) = \mathbb{E} \left[\sum_{t=1}^T \max_{\mathbf{x} \in \mathcal{X}_t^1 \times \dots \times \mathcal{X}_t^N} r_t(\mathbf{x}) - r_t(\mathbf{x}_t) \right].$$

- GLM Bandits require dealing with an (exponential) *non-linearity* parameter κ ;

$$\kappa = \max_{\mathbf{x} \in \{\mathcal{X}_t^1 \times \dots \times \mathcal{X}_t^N\}_{t \in [T]}} \max_{\boldsymbol{\theta} \in \Theta} \frac{1}{\dot{\mu}(\mathbf{x}^\top \boldsymbol{\theta})}.$$

Limited Adaptivity

The learner is constrained to make only M policy updates.

Batched Setting

- Learner has to decide the update schedule *a priori*.
- The item-sets are assumed to be sampled from an unknown $\mathcal{D}$.
- $\Theta(\log \log T)$ updates are required and suffice for optimal regret [1,4].

Rarely-Switching Setting

- The learner can choose when to update the policy *on-the-fly*.
- The item-sets are assumed to be adversarially generated.
- $\tilde{\Omega}(d \log T)$ updates are required for optimal regret [2].

Applying SOTA algorithms [4] to our setting results in inefficient algorithms.

Goals

- Achieve κ -free optimal regret under limited adaptivity constraints.
- Achieve efficiency, i.e, avoid iteration over exponentially large set of slates.
- Handle single (*bandit*) feedback.

Diversity Assumption (A1)

The items $\mathbf{x}_t^i$ chosen by our algorithm for all $t \in [T], i \in [N]$ are *diverse* enough, i.e,

$$\exists \rho > 0 \text{ s.t. } \mathbb{E}[\mathbf{x}_t^i \mathbf{x}_t^{i\top} \mid \mathcal{F}_{t-1}] \geq \rho \mathbf{I} \quad \text{and} \quad \mathbb{E}[\mathbf{x}_t^i \mid \mathcal{F}_{t-1}] = \mathbf{0}.$$

This is strictly weaker than the one used in [3], who assume $\mathbb{E}[\mathbf{x}_t^i \mathbf{x}_t^{i\top} \mid \mathcal{F}_{t-1}] \geq \rho \kappa \mathbf{I}$.

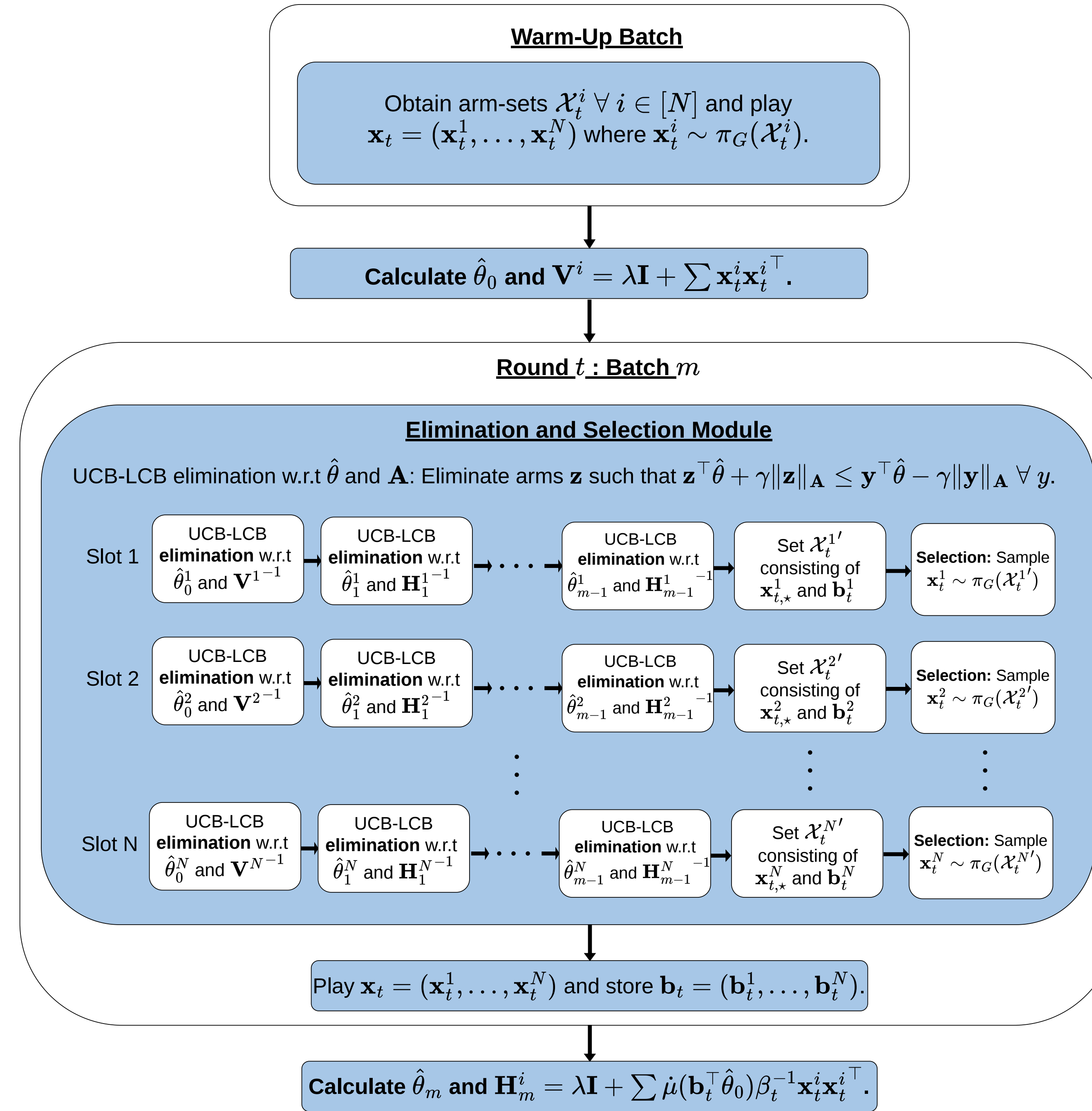
Implication of (A1)

- We show that the eigenvalues of the design matrices grow linearly with time:
 $\forall i \in [N]$, let $\mathbf{H}_t^i = \sum \dot{\mu}(\mathbf{x}_t^\top \boldsymbol{\theta}^*) \mathbf{x}_t^i \mathbf{x}_t^{i\top}$, then, $\mathbb{P}[\lambda_{\min}(\mathbf{H}_t^i) \geq c\rho t] \geq 1 - \delta$.
- This, in turn, allows us to show to show a *multiplicative equivalence*, i.e
 $\mathbf{H}^{-1} \geq c \cdot \mathbf{H}^{i-1} \implies \|\mathbf{x}\|_{\mathbf{H}^{-1}} \leq \sqrt{c} \cdot \|\mathbf{x}^i\|_{\mathbf{H}^{i-1}}$, where $\mathbf{H} = \sum \dot{\mu}(\mathbf{x}^\top \boldsymbol{\theta}^*) \mathbf{x} \mathbf{x}^\top$.

Optimal Designs

- G-Optimal Designs:** These are the solution to the optimization problem:
 $\pi_G(\mathcal{X}) = \arg \min_{\pi \in \Delta(\mathcal{X})} \max_{\mathbf{x} \in \mathcal{X}} \|\mathbf{x}\|_{\mathbf{V}_\pi}^2$, where $\mathbf{V}_\pi = \sum_{\mathbf{x} \in \mathcal{X}} \pi(\mathbf{x}) \mathbf{x} \mathbf{x}^\top$.
While finding π_G is NP-hard, approximate solutions can be found efficiently.
- Distributional Optimal Designs:** Introduced by [2], these give optimal design guarantees for stochastically sampled arm-sets.

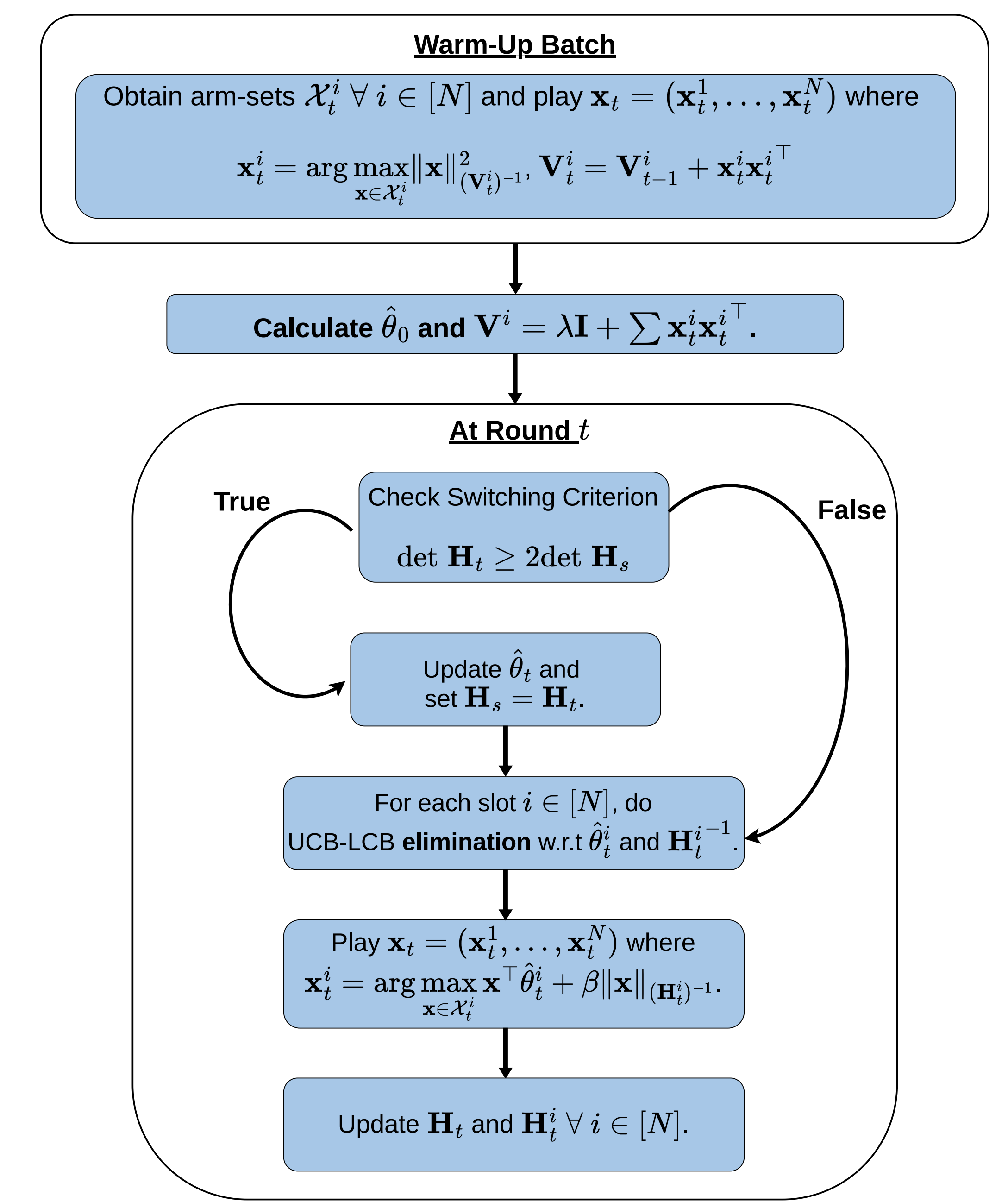
B-SlateGLinCB: A Batched Algorithm



Key Ideas and Highlights

- Idea: (A1)** allows to *reduce* the slate-level problem to a slot-level problem.
- Extending [4]:** When $N > 1$, we get the terms $\dot{\mu}(\mathbf{x}^i \mathbf{x}_t^{i\top}) / \dot{\mu}(\mathbf{x}^\top \boldsymbol{\theta}^*)$ (can be $\propto \kappa$).
- Scaling Slate $\mathbf{b}_t$:** Define $\mathbf{b}_t = (\mathbf{b}_t^1, \dots, \mathbf{b}_t^N)$ as
 $\mathbf{b}_t^i = \arg \max_{\mathbf{x} \in \mathcal{X}_t^i} \|\mathbf{x}\|_{(\mathbf{V}^i)^{-1}}$.
- The Fix:** Our scaling slate leads to the terms $\dot{\mu}(\mathbf{b}_t^\top \boldsymbol{\theta}^*) / \dot{\mu}(\mathbf{x}^\top \boldsymbol{\theta}^*)$. We show that these terms grow as $\mathcal{O}(1)$. Thus, we avoid a potential κ -dependency.

RS-SlateGLinCB: A Rarely-Switching Algorithm



Key Ideas and Highlights

- Idea:** Using **(A1)**, if a "switch" happens at round t ,
 $\det \mathbf{H}_t \geq 2 \det \mathbf{H}_s \implies \|\mathbf{x}\|_{\mathbf{H}_s^{-1}} \leq \sqrt{2} \|\mathbf{x}\|_{\mathbf{H}_t^{-1}} \leq \sqrt{2c} \|\mathbf{x}^i\|_{\mathbf{H}_t^{i-1}}$.
- Removing Switching Criterion I from [4]:** A consequence of using **(A1)** to show that eigenvalues of the design matrix $\mathbf{V}^i$ grow near-linearly.
- Improvement in #Switches over [4]:** Our algorithms makes $\tilde{\mathcal{O}}(Nd \log T)$ updates, improving over [4], which makes $\tilde{\mathcal{O}}(\kappa N^2 d^2 \log^2 T)$ updates.
- Matching the lower bound:** We match the bound in [2] upto polylog factors.

Regret Bounds

B-SlateGLinCB

$$R(T) = \tilde{\mathcal{O}} \left(Nd^{3/2} \sqrt{T \cdot \mathbb{E}_{\mathcal{X}} \dot{\mu}(\mathbf{x}_*^\top \boldsymbol{\theta}^*)} \right)$$

B-SlateGLinCB with Distributional Optimal Designs

$$R(T) = \tilde{\mathcal{O}} \left(Nd\sqrt{T} \cdot \sqrt{\min \{d \mathbb{E}_{\mathcal{X}} \dot{\mu}(\mathbf{x}_*^\top \boldsymbol{\theta}^*), N \max_{\mathcal{X}} \dot{\mu}(\mathbf{x}_*^\top \boldsymbol{\theta}^*)\}} \right)$$

RS-SlateGLinCB

$$R(T) = \tilde{\mathcal{O}} \left(Nd \sqrt{\sum_{t=1}^T \dot{\mu}(\mathbf{x}_t^\top \boldsymbol{\theta}^*)} \right)$$

Experiments and References

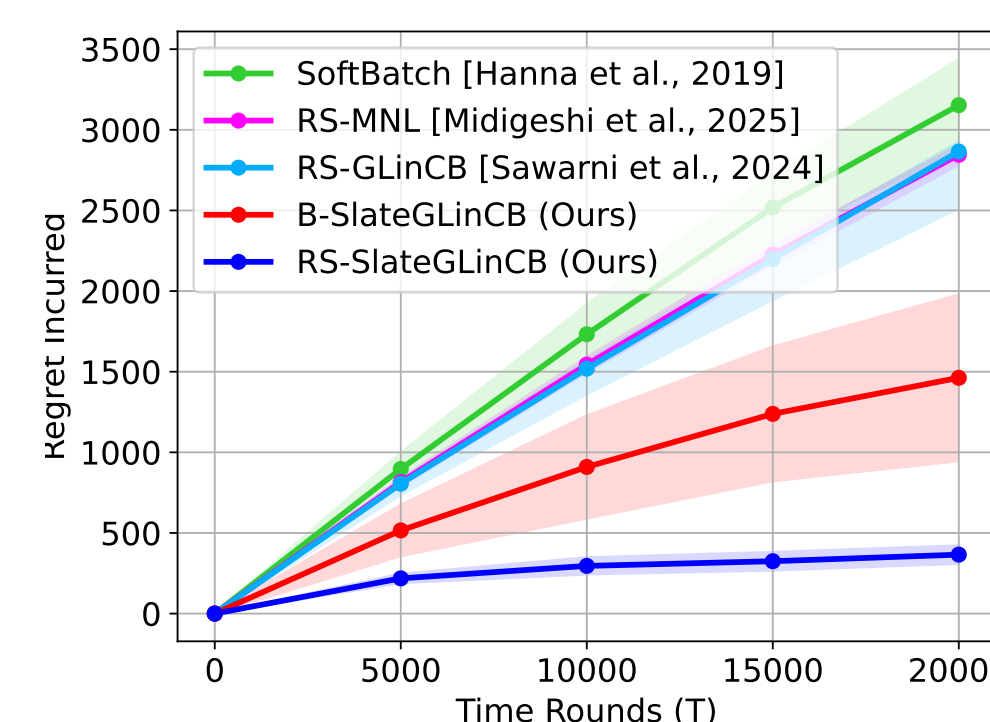


Figure 1: $\|\boldsymbol{\theta}^*\| \leq 2$, $N = 5$, $d = 5$

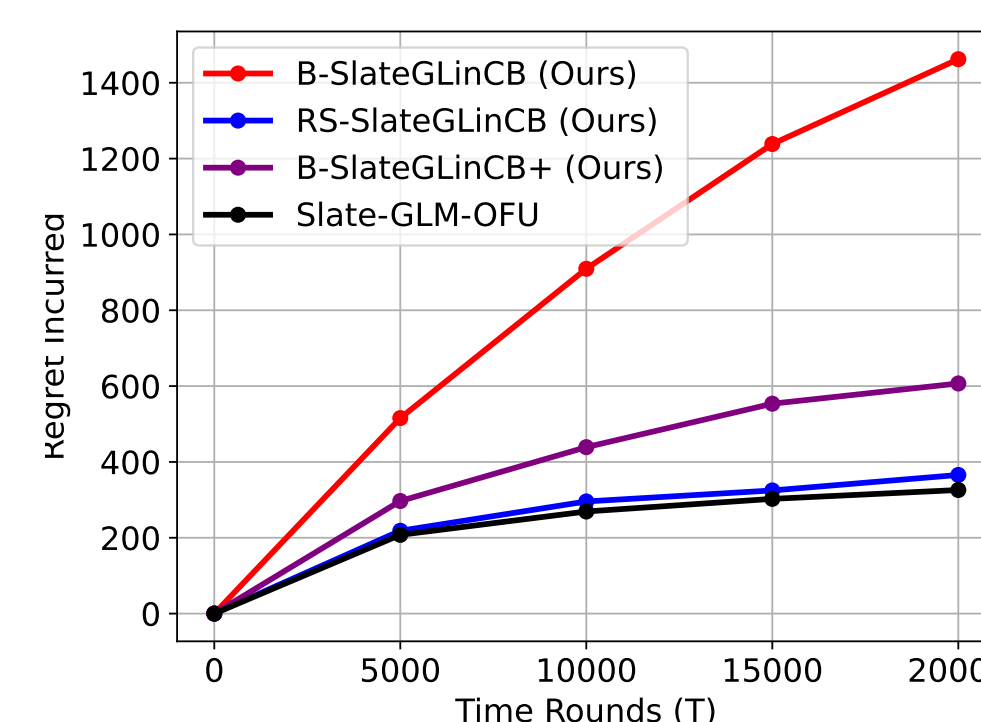


Figure 2: $\|\boldsymbol{\theta}^*\| \leq 5$, $N = 10$, $d = 5$

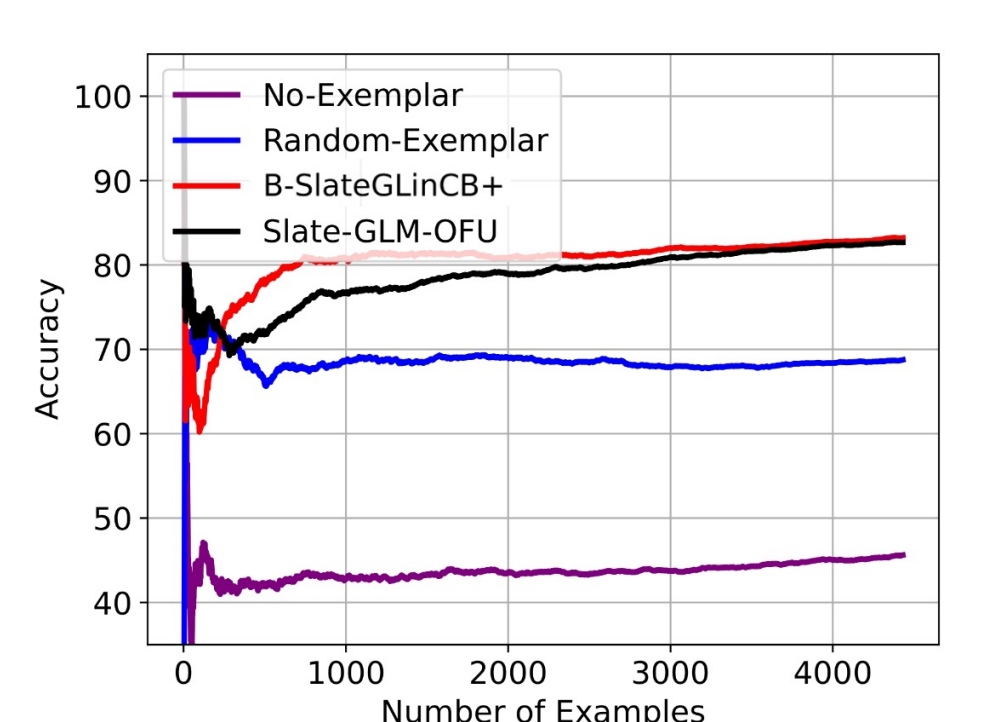
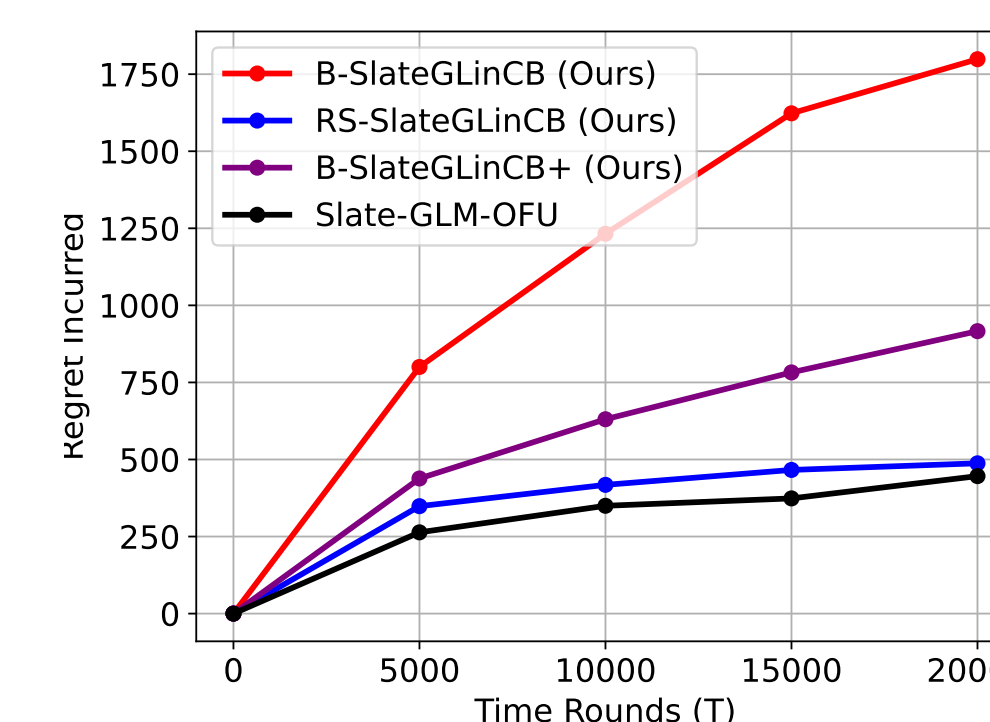
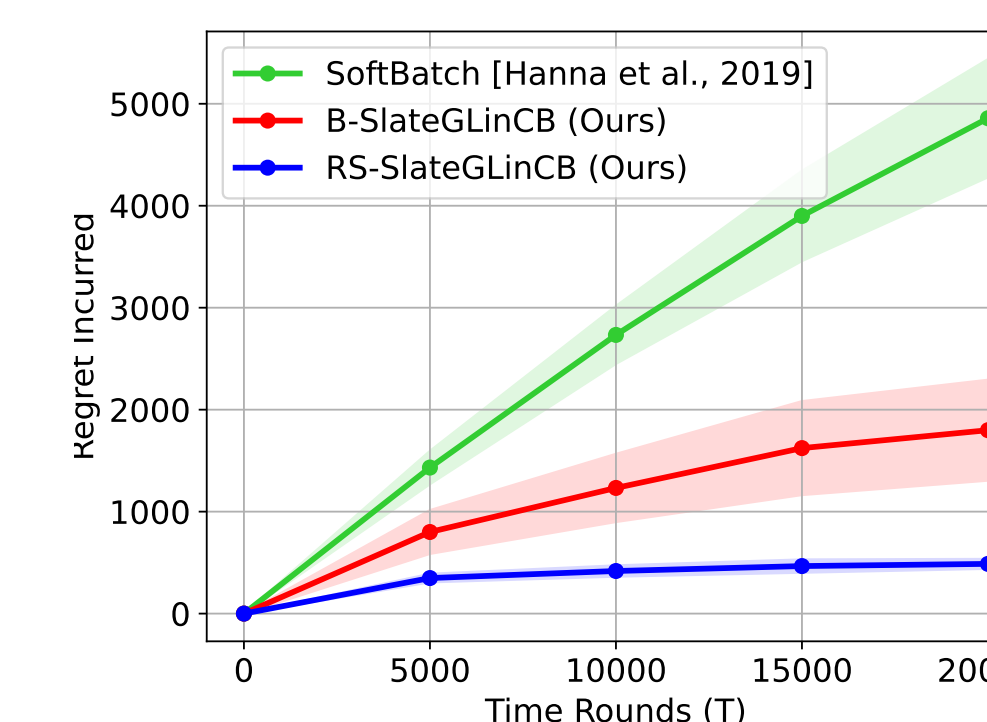


Figure 3: Prompt Tuning

[1] Gao et al. Batched Multi-armed Bandits Problem. NeurIPS 2019.

[3] Goyal and Sinha. Efficient Algorithms for Logistic Contextual Slate Bandits with Bandit Feedback. UAI 2025

[2] Ruan et al. Linear Bandits with Limited Adaptivity and Learning Distributional Optimal Design. STOC 2021.

[4] Sawarni et al. Generalized Linear Bandits with Limited Adaptivity. NeurIPS 2024.